Anuj Pai Raikar

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EDUCATION

MS Robotics Engineering, Worcester Polytechnic Institute**,** MA, USA *Aug 2022 - May 2024*

B.Tech Mechanical Engineering, Vellore Institute of Technology**,** India *July 2016 - May 2020*

SKILLS

Languages: Python, C++, MATLAB, C, SQL

Machine Learning: PyTorch, TensorFlow, Scikit-learn, ONNX, TensorRT, model evaluation, inference optimization

Robotics & Perception: OpenCV, PCL, ROS2, Nav2, Gazebo, sensor fusion, object detection, tracking, pose estimation

Systems & Tools: Linux, Docker, CUDA, Git, CI/testing, AWS, SLURM, FastAPI, NumPy, Pandas

Algorithms & Signal Processing: FFT, 2D FFT, FMCW radar simulation, range-Doppler processing, CA-CFAR detection, VAD, audio feature extraction, filtering, thresholding, statistical modeling

WORK EXPERIENCE

ML/AI Software Engineer**,** Cybic Tech Inc.**,** TX *Apr 2025 - Present*

- Architected and delivered a transformation layer for a distributed data pipeline using **FastAPI, Dagster, Trino, dbt, and SQLMesh**, enabling repeatable large-scale SQL transformations and safer rollout of new production workflows.
- Built structured execution, artifact, and validation flows for concurrent runs, including per-run isolation, configurable state handling, deterministic failure modes, and automated tests to improve reliability and debugging.
- Designed a **C++17 FFT-based audio signal gating algorithm** with interpretable signal-quality scoring, failure-mode classification, and adaptive threshold control for real-time speech inference.
- Built **statistical evaluation pipelines** using ROC curves, confusion matrices, threshold sweeps, synthetic SNR/noise tests, and feature ablations across LibriSpeech, MUSAN, and CHiME datasets.
- Developed and benchmarked **rule-based and lightweight machine learning models** for audio quality prediction, optimizing latency, memory usage, and real-time performance on NVIDIA Jetson-class hardware.

Autonomy/AI Engineer Intern**,** JLG Industries, MD *May 2024- Aug 2024*

- Built a camera-based perception system for **autonomous docking** of indoor ground robots using **PnP pose estimation** and integrated it with planning and control workflows in **ROS2, Gazebo, and Nav2**.
- Tested and debugged the perception-to-planning stack in simulation and on testbeds, achieving **5 cm docking error** across diverse trials.
- Containerized the full autonomy pipeline for reproducible deployment and system validation across development environments using **Docker**.

Robotics and ML Researcher**,** Worcester Polytechnic Institute, MA *Sep 2023 – May 2024*

Robotic Manipulation for Pick and Place Application - Manipulation & Environmental Robotics Lab

- Developed a full-stack **perception and motion planning** pipeline in ROS for autonomous object picking using a Franka robotic arm with a dexterous gripper, integrating sensor data acquisition, object segmentation, grasp planning, and execution in a real-time robotic system.
- Processed 3D point cloud data using **PCL** to extract robust grasp points, and conducted comparative benchmarking against deep learning–based approaches (**Faster R-CNN**) for grasp pose prediction, evaluating accuracy, computational efficiency, and robustness across varied object geometries.
- Implemented sampling-based motion planners **(RRTConnect and PRM)** within OMPL, integrated with a **Relaxed IK** solver to generate smooth, collision-free, and kinematically feasible end-effector trajectories under task and joint constraints.
- Designed and simulated a 3D A*-based motion planning framework for in-hand object reorientation, leveraging motion primitives and variable friction in custom dexterous gripper to try and enable precise manipulation and orientation control.

CT-to-Ultrasound Synthesis Pipeline - Medical FUSION Lab

- Developed a physics-aware neural network based ultrasound image synthesis pipeline from 3D CT volumes using **U-Net** variants with encoders enhanced by channel attention; evaluated realism using texture-based metrics (GLCM contrast, homogeneity, correlation) and speckle distribution statistics.
- Built a custom **NIfTI**-based CT data generation workflow enabling pose-conditioned ultrasound synthesis, achieving approximately **98% lower inference latency** compared to full wave-based MUST simulations.

***Machine Learning Intern*,** DEKA Research and Development, NH *May 2023 - Aug 2023*

- Developed a **Vision Transformer-based** self-supervised segmentation and anomaly detection model in **PyTorch** using pretrained DINO features, knowledge distillation, and contrastive learning.
- Built multi-GPU distributed training pipeline with monitoring and benchmarking support for large-scale experimentation.
- Benchmarked model performance, achieving **mIoU = 34** under baseline parameters and evaluating trade-offs in **accuracy vs efficiency.**

***Associate Consultant*,** Oracle Financial Services Software, India *Sep 2020 - Mar 2022*

- Implemented and supported software components across UI, database, and service layers for Oracle modules.
- Modified **JavaScript** and **PL/SQL** packages to support client-specific workflows and verified integrations through debugging, QA, and web-service validation.
- Collaborated across client and internal teams to document requirements, track issues in JIRA, and support reliable delivery in a production enterprise environment.

PROJECTS

FMCW Automotive Radar Signal Processing for Moving-Target Detection — *MATLAB, FFT, CFAR*

- Built a MATLAB-based automotive radar pipeline for target scenario generation, **FMCW waveform creation**, beat-signal processing, and **Range-Doppler Map generation**.
- Applied 1D FFT, 2D FFT, and CA-CFAR adaptive detection to estimate target range/velocity and suppress false detections under noise.
- Designed waveform parameters from radar requirements, including range resolution bandwidth, chirp timing, maximum unambiguous range, and FMCW slope.
- Used MATLAB Driving Scenario Simulator to model multi-object scenes and analyze radar clustering, detection association, and **multi-object tracking behavior**.

Reinforcement Learning based Autonomous Driving Stack - *Python, PyTorch, C++17, ONNX, TensorRT, nuPlan*

- Extended a **transformer** based **imitation learning** autonomous driving planner on **nuPlan dataset** with **a geometry-aware trajectory mode re-ranker** (custom CUDA kernel, 13.7× CPU speedup), improving closed-loop benchmark performance by 0.77% on an 84-scenario evaluation suite without retraining the base model.
- Designed a reproducible evaluation framework with scenario bucketing and targeted failure analysis (intersections, stop-and-go, pedestrians, lane changes) to surface failure modes correlating with on-road safety.
- Implemented **ONNX, TensorRT**, and native **C++** inference pipelines for a transformer-based autonomous driving planner; achieved **0.77 ms FP16** and **1.34 ms FP32 GPU latency** with cross-backend fidelity validation against **PyTorch** reference.
- Built test harnesses, CI (GitHub Actions), and automated benchmark scripts to validate inference correctness and reproducibility across Python and C++ runtimes.

LiDAR Obstacle Detection and Tracking — *C++14, PCL*

- Built a real-time **3D computer vision pipeline** for highway driving using custom **KD-tree**, **RANSAC** ground segmentation, and Euclidean clustering on KITTI point clouds.
- Applied ROI filtering, ego-vehicle removal, and voxel-grid downsampling to reduce latency by **50%** from **10 ms to 5 ms per frame**.
- Implemented **multi-object tracking** with nearest-neighbor association and constant-velocity prediction for temporally consistent 3D obstacle tracking.